

MANAGEMENT OF ACUTE AND RECURRENT BACK PAIN **IN PRIMARY CARE**

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BACKGROUND & KEY MESSAGES

- **Acute back pain** is common and lasts less than 3 months
- Pain is generally **non-specific**; precise and exact diagnosis is often not possible or necessary
- After acute episodes, there may be persistent or fluctuating pain for a few weeks to months
- Even in the presence of severe pain that limits activities initially, **this tends to improve**, with possible recurring episodes and occasional pain afterwards
- Acute back pain does not cause prolonged loss of function unlike **chronic back pain**

- **Chronic back pain** is defined as persistent pain lasting more than 3 months. (This is different from **recurrent back pain**: episodic acute back pain that lasts less than 3 months but recurs after a period with minimal / mild symptoms that do not affect activities or function)
- Patients with chronic back pain are more likely to report loss of function or activity restrictions.
- Consider specialist review for patients with chronic back pain despite initial adequate education, reassurance, analgesia and trial of physiotherapy.

Key Messages

- *Acute back pain is common. Episodes are mostly short-lived (symptoms improve within 4 weeks)², and reassurance is very helpful*
- *In the absence of red flags, investigations in the first 4 – 6 weeks do not provide clinical benefit³⁻⁵*
- *There is strengthened evidence to encourage patients to remain active: resume usual activities, including work, as soon as possible⁶⁻⁷*
- *Analgesia and physiotherapy may provide short term symptom control*
- *Some clinical interventions may be harmful, especially extended bed rest, extended use of opiates, NSAIDs or benzodiazepines*

CLINICAL ASSESSMENT

- Rule out **red flags** first: If present, early investigations, urgent referrals to A&E or spine specialist need to be considered

- Features of cauda equina syndrome (new-onset incontinence, saddle anaesthesia, lax anal tone, progressively worsening lower limb neurological symptoms)
- Weight loss, current or history of cancers
- Significant trauma or history of osteoporosis
- Fever, long term steroids use, or other forms of immunosuppressant
- Severe, unremitting night pain

A red, wavy-edged banner with the text "RED FLAGS" in white capital letters.

RED FLAGS

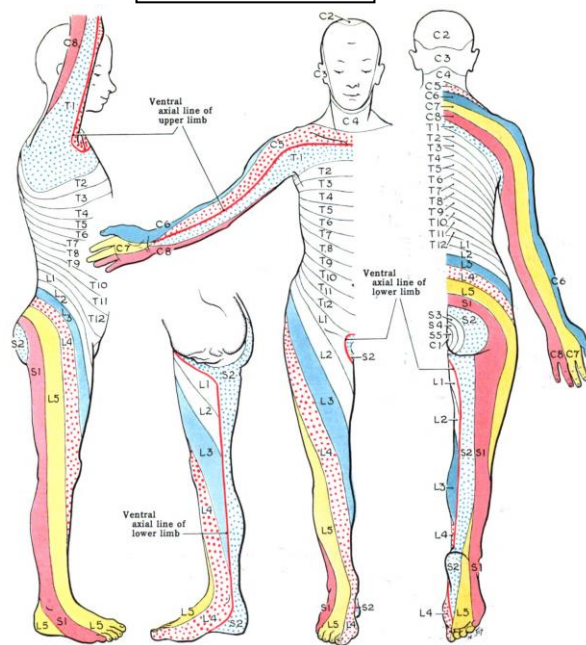
History

- History of the current acute episode (duration of pain, pain score, trigger, relieving factors and response to previous therapy *etc.*)
- History of previous similar symptoms (recurrent back pain), how current symptoms compare to the past
- Activities associated with pain
- Presence of radiculopathy or claudication
- Presence of early morning stiffness of at least an hour
- Degree of impairment on occupation and activities of daily living
- Presence of mood disorder (depression / anxiety) especially in patients with chronic lower back pain

Physical Examination

- Should be focused and targeted with special manoeuvres performed only if relevant
 - E.g. Look for dermatomal / myotomal distribution of neurological deficits when radicular pain is present
 - Palpation of SI joints if there is suspicion of SI joint pathologies
 - Consider Schober's test when history and physical findings are suggestive of restriction in lumbar flexion commonly seen in ankylosing spondylitis
- Aim to identify any relevant abnormal signs that may indicate further evaluation is required, in addition to reaching possible differential diagnoses
- Physical examinations that should always be considered
 - Inspection of back and posture: for anatomical abnormalities: scoliosis, hyperkyphosis, and loss of lumbar lordosis or listing
 - Palpation of the spine for **localised** vertebral tenderness (can suggest spinal infection, metastases or compression fracture)
 - Lower limb neurological examination: strength, reflexes, sensation and gait

Dermatomes

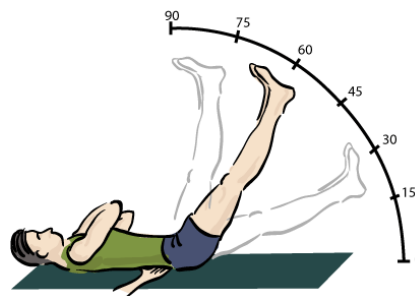


Myotomes

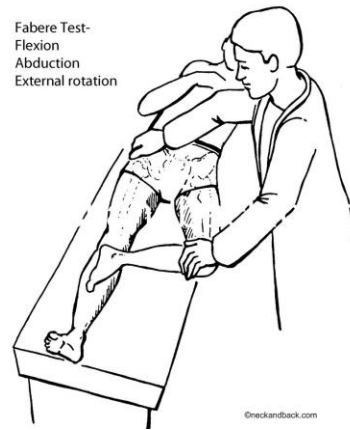
Nerve Root	Action
L2	Hip Flexion
L3	Knee Extension
L4	Ankle Dorsiflexion
L5	Big toe Extension
S1	Ankle Plantarflexion

Special manoeuvres (only if relevant):

- **Straight leg raise (SLR or Lasègue's sign):** positive when radicular pain (pain that radiates beyond the ipsilateral knee, not just localized to lower back or hamstring) occurs between 30 – 70 degrees of hip flexion

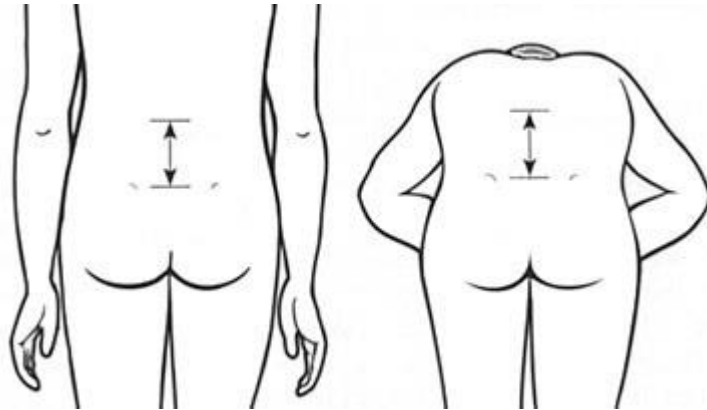


- **Patrick's (or Flexion ABduction External Rotation) test:** positive when hip or buttock pain is elicited when the ipsilateral leg is **F**lexed at the knee with the hip **AB**ducted and **E**xternally **R**otated – positive test raises suspicion of a hip or sacroiliac joint pathology
 - Examples of conditions associated with the location of pain elicited with the manoeuvre:
 - 1) Elicited at the SI joint: Sacroiliitis or SI joint dysfunction
 - 2) Elicited at the groin: Iliopsoas strain / bursitis, intra articular hip disorder
 - 3) Elicited at the posterior hip: Posterior hip impingement



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- **Schober's test** (only when indicated as it takes a longer time to perform in a busy clinic): 2 points are marked (10 cm above and 5 cm below) from the level of L5 vertebral. The patient is asked to bend forward and touch his / her toes. Distance between these 2 points should increase at least by 5 cm (to a total distance of at least 20 cm), if not restriction in lumbar flexion should be suspected.



INVESTIGATIONS

Laboratory Investigations

- Most patients with acute low back pain do not require any laboratory investigations
- Do consider etiologies outside of the spine: e.g. pancreatitis, pyelonephritis, nephrolithiasis, abdominal aortic aneurysm or herpes zoster
 - Patients generally have other accompanying symptoms
 - Relevant investigations can be done if clinical suspicion is high
- If there is suspicion of the back pain being part of a systemic illness such as connective tissue disorder, inflammatory spondyloarthropathy or infection, inflammatory markers such as **ESR** can be considered
 - Normal ESR can help to exclude suspected inflammatory arthritis/infection but are not specific if elevated

Radiological Investigations

- Earlier use (within first 4 – 6 weeks of presentation) of imaging for low back pain without **red flags** is not associated with improved outcome
- Radiological investigations often show abnormal findings in asymptomatic patients, making it difficult to correlate radiological findings with clinical symptoms
- Inappropriate lumbar imaging can lead to irrelevant findings and trigger additional costly studies, unneeded treatments, and unwarranted surgical interventions
- Only plain X-rays are available in National University Polyclinics

When plain radiographs are indicated, anteroposterior (**AP**) and lateral (**Lat**) views are usually adequate. X-ray of the SI joint can be performed if there is clinical suspicion of sacroiliitis.

INDICATIONS FOR EARLY RADIOLOGICAL INVESTIGATIONS

- Significant trauma (e.g. high energy injuries: road traffic accident, fall from height) or low energy trauma in elderly patients
- Current or **previous** history of cancers
- **Risk factors** for spinal infections (e.g. fever, IV drug abuser, immunosuppression, recent sepsis, **recent spine procedure**)
- High risks for vertebral compression fractures (e.g. advanced age, prolonged systemic steroids use, known osteoporosis)

Beware of the limitations of plain x-rays³

- May not be sensitive enough in certain clinical scenarios
- Certain patients will require urgent advanced imaging modalities (e.g. MRI)
 - Patients with cauda equina syndrome or significant progressive neurological deficits
 - Patients with very high clinical suspicion for spinal infections
 - Patients with very high clinical suspicion for malignancies
- These patients should either be referred to A&E if clinically indicated or given urgent specialist appointments for further evaluation

TREATMENT OPTIONS FOR NON-SPECIFIC LOW BACK PAIN

Non-Pharmacological

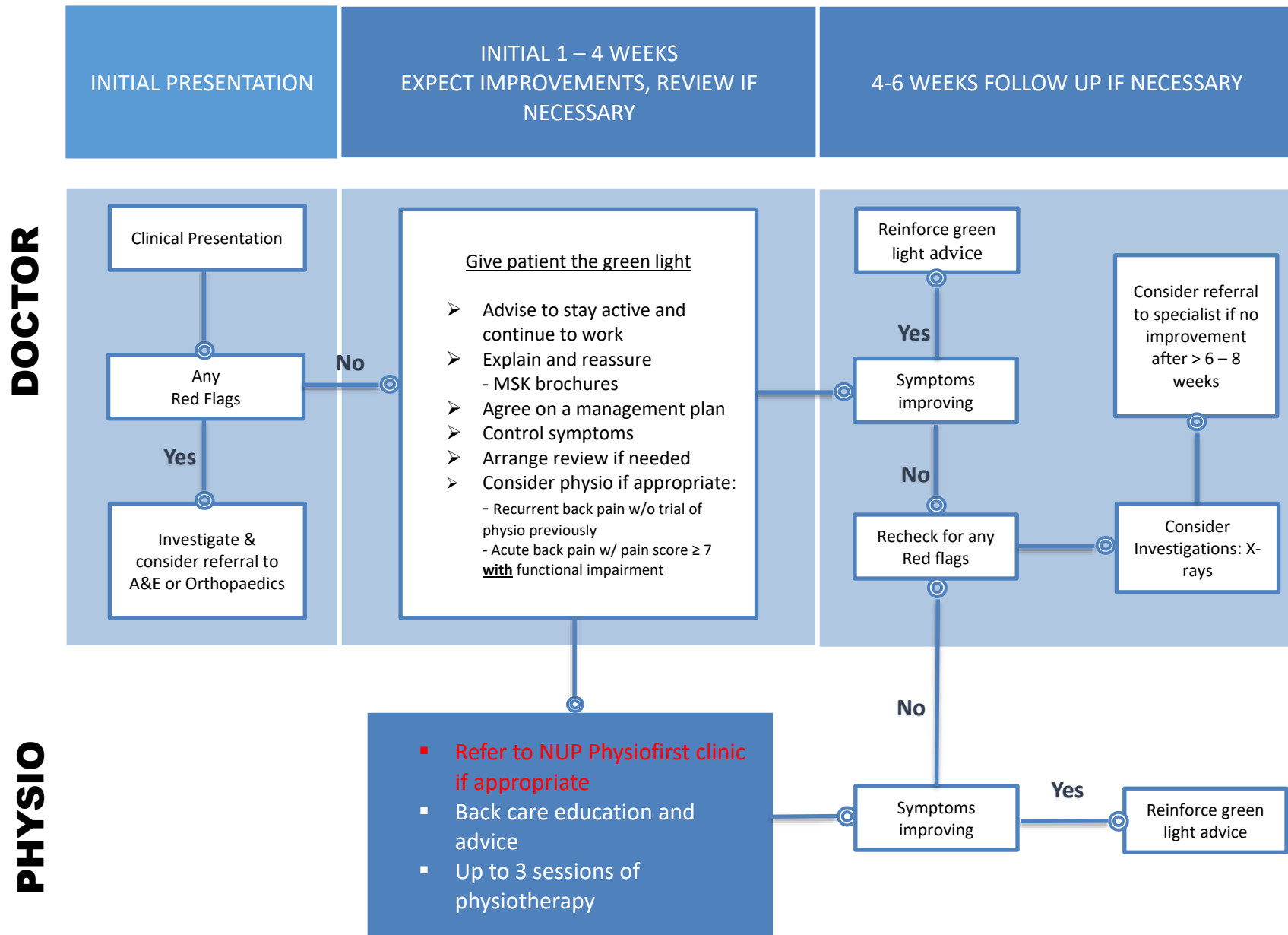
- Patient education and reassurance **is key**
 - **Reassure** patients: prognosis is often good, and most cases resolve with little intervention
 - Use information booklet / brochure (e.g. [NUP MSK Brochure](#)) to help you
 - Agree on a management plan (e.g. if no improvement return for a review, refer for physio *etc.*)
- Encourage patients to **stay active** and return to their usual activities as soon as possible
- Avoid bed rest
- Lifestyle modification: avoid twisting and bending, avoid heavy contact sports and strenuous activities that put a heavy load on the back
- Physiotherapy – consider if symptoms last more than 2 weeks despite adequate trial of analgesia

Pharmacological

- Start with a trial of 1 - 2 weeks of appropriate analgesia
 - Can be given as a regular dosing regimen rather than PRN dosing regimen initially especially in the setting of acute back pain
- **NSAIDs**
 - They have the strongest evidence for symptom relief in acute back pain
 - NSAIDs should be used with caution or avoided in patients with renal impairment (CKD 3 or worse)
 - NSAIDs should be avoided in patients taking concomitant antiplatelet / anticoagulant

- NSAIDs / selective cyclo-oxygenase inhibitors should be used with extreme caution (consider using a lower dose), or preferably avoided in the elderly even if serum creatinine is within the normal reference range
- No clear evidence that any **particular** NSAID is superior in the management of acute back pain
- Switching to a different NSAID can be considered if the first is ineffective
- **Paracetamol with or without Orphenadrine**
 - Paracetamol with or without Orphenadrine may be used as first line analgesia
 - It is a safer first line options for patients with contraindication to NSAIDs (asthmatic, CKD patients *etc.*)
 - Beware of sedation risks with Orphenadrine and adverse effect on elderly patients
 - Paracetamol may be offered in patients who are unable to tolerate Orphenadrine or NSAIDs
 - Evidence for this group of analgesia **is weaker** than NSAIDs in acute back pain
- **Opioids / Tramadol**
 - Should only be used as an alternative (when other medications are contraindicated)
 - Their use if necessary should be limited to only a few days and not more than 2 weeks
 - Be aware of the potential risks for abuse
- **Antiepileptics**
 - Usually used for radicular pain although evidence supporting their use is limited
 - Gabapentin / Pregabalin have very low-level evidence for use in chronic radicular pain
- **Topical analgesia**
 - Low level evidence for the use of topical capsaicin
 - Very little evidence for the use of other topical analgesics

RECOMMENDED MANAGEMENT ALGORITHM FOR ACUTE BACK PAIN



RECURRENT AND CHRONIC BACK PAIN

- Patients who responded to conservative treatment for their acute back pain can have periods with no / minimal symptoms that do not interfere with their daily activities before another episode of back pain
- For patients with **recurrent back pain**, it is important to revisit the history and physical examination, to ensure there is no onset of new symptoms or physical findings that can suggest a more serious etiology (including the presence of **red flags**)
- Approach to the management of a recurring episode of back pain (in the absence of any worsening or new worrying symptoms) that is **less than 3 months** is similar to that of acute back pain, with conservative management taking a central role
- Consider offering physiotherapy if not already done so
- Consider referring to a specialist (spine team) **if**:
 - There is **chronic back pain** (persistent back pain without any symptom-free periods) for more than 3 months despite appropriate conservative management
 - Presence of **worsening** symptoms or **RED FLAGS**
 - Patient's function is significantly impaired (frequent absence from work, daily activities of living affected *etc.*)
 - Recurrent back pain that is not well controlled with conservative management
- Patients with chronic pain are also more likely to have underlying mood disorders (anxiety, depression), consider screening for them if appropriate

REFERRAL TO A SPINE SPECIALIST

- The majority of patients with acute back pain **do not need** to be referred to a specialist
- Certain patients should be referred immediately to A&E
 - New onset cauda equina syndrome **symptoms**
 - High suspicion of spinal infection (**febrile, septic , significant rest pain, spinal tenderness, risk factors**)
 - High energy injury with spinal fractures (road traffic accident, fall from height *etc.*)
- Some patients should be referred urgently to a spine specialist
 - High suspicion of **primary spinal malignancy** or metastatic diseases (including pathological fractures)
 - Compression fractures with **persistent** significant pain
 - **Severe unremitting or worsening radicular symptoms with history and examinations suggestive of a prolapsed intervertebral disc (PID), particularly those with any neurological deficit**
 - Patients with milder stable symptoms may not need to be referred to spine specialist immediately
 - They may have a trial of conservative management first
- Patients with recurrent or chronic back pain that does not respond to adequate conservative management can be referred
 - They should have a trial of physiotherapy **if not previously done** after red flags are excluded

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